

Ideation Phase - I

Brainstorm & Idea Prioritization

Date	01 March 2026
Team ID	Team 154
Project Name	Insurance Fraud Detection using ML
Maximum Marks	4 Marks

Step 1: Team Gathering, Collaboration and Problem Statement

Problem Background

Insurance companies lose **billions of dollars every year due to fraudulent claims**. Fraudsters manipulate claim information such as accident details, medical bills, or claim history to obtain illegal payments.

Traditional fraud detection methods rely on **manual investigation**, which is:

- Slow
- Expensive
- Inefficient for large datasets

With the growth of digital insurance platforms, there is a need for **automated systems** that can detect fraudulent claims quickly and accurately.

Problem Statement

Develop a **Machine Learning based system** that analyzes insurance claim data and predicts whether a claim is **fraudulent or legitimate**.

The system should:

- Analyze claim features such as claim amount, customer history, accident details, etc.
 - Use machine learning algorithms to detect suspicious patterns.
 - Provide fast predictions to assist insurance companies in investigation.
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Objectives

1. Detect fraudulent insurance claims automatically.
 2. Reduce financial losses for insurance companies.
 3. Improve accuracy compared to manual fraud detection.
 4. Provide a real-time prediction system via a web API.
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Stakeholders

- Insurance companies
 - Fraud investigation teams
 - Insurance customers
 - Data analysts
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Step 2: Brainstorm, Idea Listing and Grouping

During the brainstorming session, the team discussed multiple ideas related to fraud detection and how machine learning can solve the problem.

Brainstormed Ideas

Idea 1: Rule-Based Fraud Detection

Use predefined rules such as:

- Claim amount exceeding threshold
- Multiple claims within short period

Limitation: Not flexible and easily bypassed.

Idea 2: Machine Learning Classification Model

Train models using historical claim data to classify claims as **Fraud or Legitimate**.

Possible Algorithms:

- Logistic Regression
- Random Forest

- Decision Tree
 - Gradient Boosting
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Idea 3: Anomaly Detection

Detect unusual patterns in claims that differ from normal claims.

Techniques:

- Isolation Forest
 - Autoencoders
 - Outlier detection
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Idea 4: Behavioral Analysis

Analyze customer behavior such as:

- Claim frequency
 - Claim timing
 - Repeated suspicious patterns
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Idea 5: Real-Time Fraud Prediction API

Deploy a model using **Flask API** so insurance companies can send claim data and get predictions instantly.

Step 3: Idea Prioritization

Selected Solution

The final selected solution is:

Machine Learning Based Fraud Detection System

The system will:

1. Use historical insurance claim data.
 2. Train a classification model to detect fraud.
 3. Deploy the trained model using a **Flask API**.
 4. Allow users to send claim data and receive fraud predictions.
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Expected Outcomes

- Accurate detection of fraudulent insurance claims
 - Faster fraud investigation
 - Reduced financial losses
 - Scalable fraud detection system
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